

Dynamic Inconsistency in Dual-Shock Monetary Policy: A Post-COVID Framework

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Abstract

This paper extends the classic Kydland-Prescott dynamic inconsistency framework to environments in which central banks simultaneously face supply and demand shocks with conflicting optimal policy prescriptions. We develop a dual-shock Phillips curve and a modified Taylor rule that disentangle inflationary pressures by source, introducing asymmetry and nonlinearity into the central bank's optimization problem. The model shows that traditional monetary policy frameworks become dynamically inconsistent when overlapping shocks occur—particularly under imperfect information about shock composition. We characterize equilibrium under both commitment and discretion, discuss welfare implications, and argue that the time-inconsistency problem is amplified by learning effects, correlation surprises, and evolving uncertainty. The framework reveals new sources of inflation bias and welfare losses even in otherwise stable macro environments. These theoretical results offer a foundation for rethinking rule-based frameworks and policy communication in an era of heightened supply-side volatility and asymmetric

trade-offs. A stylized discussion of recent monetary policy dilemmas illustrates the practical relevance of the model’s mechanisms.

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1 Introduction

The post-COVID inflation surge exposed fundamental weaknesses in traditional monetary policy frameworks. Between 2021 and 2022, the Federal Reserve struggled to distinguish between supply chain disruptions and demand pressures, initially characterizing inflation as “transitory” before implementing the fastest rate hiking cycle since the 1980s. This policy reversal highlights a critical gap in monetary theory: existing frameworks inadequately address dynamic inconsistency problems when central banks face simultaneous supply and demand shocks requiring conflicting optimal responses.

The classic Kydland and Prescott (1977) time-inconsistency problem assumes central banks face a standard inflation-unemployment trade-off. However, modern economies experience complex shock structures where supply disruptions (trade wars, pandemics, geopolitical events) create inflationary pressures without output expansion, while demand shocks (fiscal stimulus, credit expansion) move inflation and output in the same direction. When these shocks occur simultaneously, central banks face enhanced dynamic inconsistency problems as optimal policy responses differ fundamentally across shock types.

This paper makes three key contributions. First, we extend the Kydland-Prescott framework to dual-shock environments, showing how shock uncertainty amplifies the time-inconsistency problem. Second, we develop a dual-shock Phillips curve that separates supply from demand pressures and propose a modified Taylor rule with a threshold response to supply shocks. Third, we discuss how the Federal Reserve’s 2020-2024 experience illustrates these the-

oretical predictions, with shock misidentification contributing to significant policy mistakes.

Our theoretical analysis suggests that traditional Taylor rules perform poorly when supply shocks are large or when shocks are incorrectly identified. The welfare costs of discretionary policy increase substantially under dual shocks, particularly when supply and demand pressures are correlated. These findings have important implications for central bank design, communication strategies, and policy rules in an era of increased economic fragmentation and supply chain vulnerability.

The paper proceeds as follows. Section 2 establishes the theoretical framework, extending standard models to dual-shock environments. Section 3 develops the mathematical model with proof sketches of the key propositions. Section 4 analyzes the enhanced dynamic inconsistency problem under shock uncertainty. Section 5 discusses Federal Reserve policy during 2020-2024 through the lens of the framework. Section 6 concludes with policy implications for modern central banking.

2 Theoretical Framework

2.1 Basic Setup and Key Innovations

We extend the Kydland and Prescott (1977) framework to incorporate two distinct shock processes affecting the economy. Following Clarida, Galí, and Gertler (2000) and Woodford (2003), we take the New Keynesian policy framework as our starting point, but we work with a Phillips curve in which expectations are formed one period in advance and augment it with separate treatment of supply and demand disturbances.

Economic Structure: The economy features three key relationships. The dual-shock Phillips curve separates inflation pressures by source:

$$\pi_t = \alpha \pi_t^e + \beta_D \max(D_t, 0) + \beta_S S_t + \varepsilon_t \quad (1)$$

where π_t is inflation, π_t^e denotes inflation expectations, D_t represents demand shocks with asymmetric effects (only positive shocks matter), S_t captures supply shocks (affecting inflation regardless of sign), and ε_t is white noise.

The output gap evolves according to:

$$y_t = \phi_D D_t - \phi_S |S_t| + \eta_t \quad (2)$$

where demand shocks boost output while supply disruptions reduce it. The monetary policy instrument follows a modified Taylor (1993) rule:

$$r_t = r^* + \phi_\pi(\pi_t - \pi^*) + \phi_y y_t + \phi_S f(S_t) + u_t \quad (3)$$

where $f(S_t) = S_t \cdot \mathbf{1}(|S_t| > \bar{S})$ represents threshold-based supply shock responses and $\bar{S} > 0$ is the response threshold.

Central Bank Preferences: The monetary authority minimizes expected losses over both inflation and output deviations, with an additional shock-dependent cost term reflecting the differential nature of shock responses:

$$L_t = \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s [\alpha(\pi_{t+s} - \pi^*)^2 + \lambda_D (y_{t+s})^2 + \lambda_S \omega(S_{t+s}) + \gamma(\Delta\pi_{t+s})^2] \quad (4)$$

where $\omega(S_t) = \exp(\kappa|S_t|) - 1$ captures non-linear costs of supply disruptions and $\gamma \geq 0$ represents preferences for inflation smoothing.

Key Innovation: Unlike standard models that treat all shocks symmetrically, our framework recognizes that supply shocks create inflation-output trade-offs (Blanchard and Galí, 2007) while demand shocks generate positive correlation between inflation and output gaps. This distinction fundamentally alters optimal policy design and creates new sources of time-inconsistency.

2.2 Shock Processes and Information Structure

Both shock processes follow persistent AR(1) dynamics:

$$D_t = \rho_D D_{t-1} + \epsilon_{D,t}, \quad \epsilon_{D,t} \sim N(0, \sigma_D^2) \quad (5)$$

$$S_t = \rho_S S_{t-1} + \epsilon_{S,t}, \quad \epsilon_{S,t} \sim N(0, \sigma_S^2) \quad (6)$$

with potential correlation $\mathbb{E}[\epsilon_{D,t}\epsilon_{S,t}] = \sigma_{DS}$.

Crucially, the central bank observes aggregate economic outcomes but faces a signal extraction problem in distinguishing shock sources. This information friction creates additional dynamic inconsistency as policy commitments based on expected shock compositions become suboptimal when actual realizations differ from expectations.

3 The Dual-Shock Model

3.1 Equilibrium Characterization

Definition 1. A rational expectations equilibrium consists of sequences

$$\{\pi_t, y_t, r_t, \pi_t^e\}_{t=0}^{\infty}$$

such that:

1. The Phillips curve (1) and output equation (2) hold;
2. The central bank chooses r_t optimally, either period by period taking π_t^e as given (discretion) or once and for all (commitment);
3. Expectations are rational: $\pi_t^e = \mathbb{E}_{t-1}[\pi_t]$.

Proposition 1 (Equilibrium Existence): A unique rational expectations equilibrium exists if:

1. $|\alpha + \phi_\pi \beta_D \phi_D| < 1$ (determinacy condition);

2. $\sigma_D^2 + \sigma_S^2 < \infty$ (finite variances);
3. $|\rho_D|, |\rho_S| < 1$ (stationary shock processes).

Proof Sketch: To establish existence and uniqueness of a rational expectations equilibrium, we transform the system into a **state-space representation** and apply the **Blanchard-Kahn conditions**.

First, define the vector of predetermined state variables and forward-looking variables. Let $\mathbf{x}_t = [D_t, S_t]'$ be the vector of predetermined state variables, and π_t be the key endogenous variable whose expectations influence current outcomes. The core equations of the model are:

$$\begin{aligned}\pi_t &= \alpha\pi_t^e + \beta_D \max(D_t, 0) + \beta_S S_t + \varepsilon_t \\ y_t &= \phi_D D_t - \phi_S |S_t| + \eta_t \\ D_t &= \rho_D D_{t-1} + \epsilon_{D,t} \\ S_t &= \rho_S S_{t-1} + \epsilon_{S,t}\end{aligned}$$

Under discretion, the central bank chooses r_t taking π_t^e as given. The optimal π_t (from equation (8) below) and y_t (from (2)) define the relationships between current shocks and outcomes. Let $\pi_t^e = \mathbb{E}_{t-1}[\pi_t]$. Substituting this into the Phillips curve (or the central bank's optimal rule), we obtain a linear difference equation in π_t and its expectations.

The full system can be expressed in a canonical linear rational expectations form. The functions $\max(D_t, 0)$ and $|S_t|$ have a kink at zero, and $\omega(S_t)$ inherits it, so they cannot be linearized around a zero steady state. To obtain a linear system for which Blanchard-Kahn applies, we therefore consider regimes in which the model is linear ($D_t > 0$ and S_t of constant sign, with $\omega(S_t)$ approximated around a point away from zero). Within such a regime the system can be written as:

$$\mathbb{E}_t \mathbf{x}_{t+1} = \mathbf{F} \mathbf{x}_t + \mathbf{G} \mathbf{c}_t + \boldsymbol{\epsilon}_t$$

where $\mathbf{x}_t$ includes predetermined variables (like D_t, S_t) and potentially expec-

tational variables, and $\mathbf{c}_t$ are controls. A more common representation for macroeconomic models under rational expectations is:

$$\begin{pmatrix} \mathbf{x}_{t+1} \\ \mathbb{E}_t \mathbf{p}_{t+1} \end{pmatrix} = \mathbf{A} \begin{pmatrix} \mathbf{x}_t \\ \mathbf{p}_t \end{pmatrix} + \begin{pmatrix} \mathbf{B}_1 \\ \mathbf{B}_2 \end{pmatrix} \mathbf{w}_{t+1}$$

where $\mathbf{x}_t$ are predetermined variables, $\mathbf{p}_t$ are non-predetermined (jump) variables, and $\mathbf{w}_{t+1}$ collects the exogenous innovations. The **Blanchard-Kahn conditions** (Blanchard and Kahn, 1980) state that for a unique stable solution, the number of eigenvalues of the system's transition matrix that lie outside the unit circle must equal the number of non-predetermined variables.

In our system, within a linear regime, the state vector will typically include (D_t, S_t) and any endogenous state variables that emerge from the central bank's problem (e.g., if there's a lagged inflation target). The non-predetermined variable is π_t . The condition $|\alpha + \phi_\pi \beta_D \phi_D| < 1$ is the determinacy condition imposed in Proposition 1; it is intended to ensure a unique stable path for inflation expectations, ruling out explosive or indeterminate outcomes. It specifically relates to how interest rates respond to inflation (via ϕ_π) and how demand shocks affect inflation (via $\beta_D \phi_D$). If this condition is met, the system has exactly one unstable eigenvalue corresponding to the non-predetermined variable, thus ensuring a unique rational expectations equilibrium. Finite shock variances and stationarity of the shock processes are standard requirements for well-behaved stochastic processes in linear rational expectations models, ensuring that variances and expected values are finite and do not explode over time. $\square$

3.2 Optimal Policy Under Commitment

Under commitment, the central bank chooses the entire sequence $\{\pi_t\}_{t=0}^\infty$ to minimize expected discounted losses subject to private sector constraints (Phillips curve, output gap, and rational expectations). The first-order conditions yield:

$$\pi_t^C = \frac{\alpha\pi^* + \frac{\lambda_D\phi_D}{\beta_D}D_t + \frac{\lambda_S}{\beta_S}\frac{\partial\omega(S_t)}{\partial S_t}S_t}{\alpha + \frac{\lambda_D\phi_D^2}{\beta_D^2} + \frac{\lambda_S}{\beta_S^2}\left(\frac{\partial\omega(S_t)}{\partial S_t}\right)^2} \quad (7)$$

Proposition 2 (Optimal Commitment Response): Under commitment, the optimal response to shocks satisfies:

1. $\frac{\partial\pi_t^C}{\partial D_t} > 0$ if $\lambda_D > 0$ (accommodate positive demand shocks);
2. $\frac{\partial\pi_t^C}{\partial S_t} < 0$ if $\beta_S > 0, \lambda_S > 0$ (lean against supply shocks);
3. The response magnitudes depend on shock persistence through expectation formation.

Proof Sketch: The proof involves setting up a Lagrangian for the intertemporal optimization problem of the central bank, with the Phillips curve and output gap equations as constraints, along with the rational expectations condition. The first-order conditions with respect to inflation and other choice variables will yield an optimal inflation path. For condition 1, differentiating π_t^C with respect to D_t from equation (7), assuming $\lambda_D > 0$ and other parameters are positive, directly shows a positive relationship. For condition 2, differentiate π_t^C with respect to S_t . Since $\omega(S_t) = \exp(\kappa|S_t|) - 1$, then $\frac{\partial\omega(S_t)}{\partial S_t} = \kappa \exp(\kappa|S_t|) \cdot \text{sgn}(S_t)$ for $S_t \neq 0$. The term $\frac{\lambda_S}{\beta_S}\frac{\partial\omega(S_t)}{\partial S_t}S_t$ equals $\frac{\lambda_S}{\beta_S}\kappa|S_t|\exp(\kappa|S_t|)$, which is non-negative for either sign of S_t . The overall effect on π_t^C also depends on the denominator of equation (7); the full derivative is not worked out here, so the “leaning against” sign in condition 2 is not established by this sketch. Condition 3 arises because commitment policy incorporates the impact of current policy on future expectations, which depend on the persistence of shocks (ρ_D, ρ_S). This typically leads to a more aggressive response compared to discretion, as the central bank can influence expectations to a greater extent. $\square$

3.3 Discretionary Policy Solution

Under discretion, the central bank reoptimizes each period taking expectations as given. The period-by-period optimization yields:

$$\pi_t^D = \frac{\alpha\pi^* + \lambda_D\phi_D D_t + \lambda_S \frac{\partial\omega(S_t)}{\partial S_t} S_t}{\alpha + \lambda_D\phi_D^2 + \lambda_S \left(\frac{\partial\omega(S_t)}{\partial S_t}\right)^2} \quad (8)$$

Proposition 3 (Discretion vs Commitment): The discretionary equilibrium features:

1. Higher average inflation: $\mathbb{E}[\pi_t^D] > \mathbb{E}[\pi_t^C]$;
2. Suboptimal shock responses: $|\frac{\partial\pi_t^D}{\partial S_t}| < |\frac{\partial\pi_t^C}{\partial S_t}|$;
3. Welfare losses increasing in shock variances and correlations.

Proof Sketch: The argument is heuristic. The rankings in items 1 and 2 are not derived from equations (7) and (8), and under those expressions they do not hold for all parameter values: they depend on the Phillips-curve slopes β_D and β_S . They are stated here as conjectures pending a re-derivation.

1. Higher Average Inflation: This is the classic inflation-bias result of Kydland and Prescott (1977) and Barro and Gordon (1983). Under discretion, the central bank faces an incentive to create surprise inflation to stimulate output, knowing that private agents have already formed expectations. In our dual-shock model, this inflation bias persists because the Phillips curve still offers an exploitable trade-off in the short run. When π_t^e is taken as given, the central bank chooses a higher π_t to raise y_t towards its optimal level (if the output gap is negative) or to accommodate demand. The $\max(D_t, 0)$ and $|S_t|$ terms introduce non-linearities, but the fundamental inflation bias remains when π_t^e is fixed.

2. Suboptimal Shock Responses: The difference between commitment and discretion stems from the central bank's ability to influence expectations. Under commitment, the central bank internalizes the effect of its current actions on future expectations, leading to more aggressive responses that anchor inflation and mitigate the impact of shocks over time. Under discretion,

because expectations are taken as given, the central bank only reacts to the immediate shock, leading to a less effective “leaning against” for supply shocks. Mathematically, this is seen by comparing the denominators of equations (7) and (8) and how they relate to the marginal costs of inflation.

3. Welfare Losses: The welfare loss arises directly from the inflation bias and suboptimal shock stabilization. The terms in ΔW (equation (9)) show that higher shock variances (like σ_D^2 , σ_S^2) lead to greater fluctuations in π_t and y_t , which are then magnified by the suboptimal discretionary policy. The cross terms highlight that correlation between shocks can further exacerbate the problem by creating more complex policy dilemmas that a discretionary central bank is ill-equipped to handle optimally. $\square$

3.4 Model Assumptions and Robustness

This subsection explicitly addresses key assumptions and discusses the sensitivity of our theoretical results to alternative specifications, crucial for a robust mathematical model.

Treatment of Non-linearities: Our analysis in Proposition 1 (Equilibrium Existence) implicitly relies on a regime-specific treatment of the non-linear terms $\max(D_t, 0)$, $|S_t|$, and $\omega(S_t) = \exp(\kappa|S_t|) - 1$ in order to apply standard linear rational expectations techniques. The first two are piecewise linear with a kink at zero, and $\omega(S_t)$ inherits that kink, so none of them can be linearized around a zero steady state. While this provides tractability, the qualitative features of our results, particularly the enhanced dynamic inconsistency and the differential response to shock types, are expected to persist under more complex non-linear solution methods (e.g., perturbation methods, global solutions) but with increased computational complexity. The specific magnitudes of responses and welfare losses would, however, be sensitive to the degree and nature of these non-linearities.

Asymmetry of Demand Shocks: The $\max(D_t, 0)$ term in the Phillips curve implies that only positive demand shocks create inflationary pressure. If demand shocks were symmetric in their inflationary impact, the optimal policy

response to negative demand shocks would also change, potentially leading to different forms of time-inconsistency or stabilization challenges. The current specification highlights a specific policy dilemma relevant to excess demand scenarios.

Functional Form of Supply Shock Costs ($\omega(S_t)$): The exponential form $\omega(S_t) = \exp(\kappa|S_t|) - 1$ implies that the social cost of supply disruptions increases more than proportionally with the magnitude of the shock. This choice is crucial for the strong welfare implications presented in Proposition 4. If $\omega(S_t)$ were, for instance, quadratic in S_t , i.e., $\omega(S_t) = cS_t^2$, the social costs would still rise with shock magnitude, but the marginal cost would be linear rather than exponential. This would reduce the convexity of the central bank’s loss function with respect to supply shocks and consequently, while dynamic inconsistency would still exist, its magnification due to non-linearity would be less pronounced. The core insight that supply shocks create distinct policy challenges remains, but the quantitative welfare costs and the intensity of the “leaning against” behavior would be affected.

Information Structure (Shock Identification): Our model assumes the central bank faces a signal extraction problem, observing aggregate outcomes but not directly observing D_t and S_t . The specific nature of this signal extraction problem (e.g., noise-to-signal ratios, observability of different aggregates) can significantly impact the central bank’s ability to identify shocks and thus influence the magnitude of the enhanced dynamic inconsistency problem (Theorem 1). A simpler assumption of perfect shock identification would eliminate the “Learning effects” and “Correlation surprises” aspects of the enhanced time-inconsistency, though the fundamental conflict from differing optimal responses would remain.

Parameter Restrictions: The determinacy condition $|\alpha + \phi_\pi\beta_D\phi_D| < 1$ is essential for a unique equilibrium. Variations in parameters such as the inflation coefficient in the Taylor rule (ϕ_π), the Phillips curve slope for demand shocks (β_D), or the output gap elasticity to demand shocks (ϕ_D) can push the system towards indeterminacy or explosive behavior, requiring careful consideration of the parameter space for the model to be well-behaved. The values

of ρ_D, ρ_S close to 1 would imply very persistent shocks, making stabilization more challenging and potentially exacerbating time-inconsistency issues.

3.5 The State-Space Representation

To further support Proposition 1 and for a more complete mathematical characterization, we explicitly define the state-space representation of the model within a linear regime. Let the vector of predetermined state variables be $\mathbf{x}_t = [D_t, S_t]'$. Let the vector of jump (forward-looking) variables be $\mathbf{p}_t = [\pi_t]'$. The shock processes are:

$$\begin{pmatrix} D_t \\ S_t \end{pmatrix} = \begin{pmatrix} \rho_D & 0 \\ 0 & \rho_S \end{pmatrix} \begin{pmatrix} D_{t-1} \\ S_{t-1} \end{pmatrix} + \begin{pmatrix} \epsilon_{D,t} \\ \epsilon_{S,t} \end{pmatrix}$$

Or, more compactly: $\mathbf{x}_t = \boldsymbol{\rho}\mathbf{x}_{t-1} + \boldsymbol{\epsilon}_t$, with $\boldsymbol{\rho} = \text{diag}(\rho_D, \rho_S)$.

Within the regime $D_t > 0$, the Phillips curve is linear (omitting the white-noise term ε_t):

$$\pi_t = \alpha \mathbb{E}_{t-1}[\pi_t] + \beta_D D_t + \beta_S S_t$$

The output gap is piecewise linear in S_t (again omitting the noise term):

$$y_t = \begin{cases} \phi_D D_t - \phi_S S_t & \text{if } S_t \geq 0, \\ \phi_D D_t + \phi_S S_t & \text{if } S_t < 0. \end{cases}$$

And the policy rule in the regime $|S_t| > \bar{S}$, omitting the policy shock:

$$r_t = r^* + \phi_\pi(\pi_t - \pi^*) + \phi_y y_t + \phi_S S_t$$

A full state-space derivation would substitute y_t into the central bank's loss function, derive the reaction function, and substitute back into the Phillips curve to obtain the final system relating π_t to $\mathbb{E}_{t-1}[\pi_t]$ and the shocks. That derivation is not carried out here.

The general linear rational expectations form typically involves equations

of the form:

$$\mathbb{E}_t \mathbf{v}_{t+1} = \mathbf{A} \mathbf{v}_t + \mathbf{B} \mathbf{w}_t$$

where $\mathbf{v}_t = [\mathbf{x}'_t, \mathbf{p}'_t]'$ contains both predetermined and jump variables. Our specific system, in its structural form, relates expectations of π_t to current states. This can be rearranged into a form solvable by standard methods. The crucial aspect is that the number of unstable eigenvalues of the system's transition matrix must equal the number of jump variables (here, typically one, π_t). The determinacy condition in Proposition 1 is intended to ensure this.

4 Dynamic Inconsistency Under Shock Uncertainty

4.1 The Enhanced Time-Inconsistency Problem

The dual-shock environment creates multiple sources of dynamic inconsistency beyond the standard Kydland-Prescott mechanism.

Theorem 1 (Enhanced Dynamic Inconsistency). *In the dual-shock framework, policy chosen at $t = 0$ is time-inconsistent if any of the following hold:*

1. **Shock uncertainty:** $\text{Var}_0(S_t) \neq \text{Var}_t(S_t)$ for $t > 0$ due to learning or evolving information sets;
2. **Learning effects:** $\mathbb{E}[D_t \mid \Omega_0] \neq \mathbb{E}[D_t \mid \Omega_t]$ where Ω_t is period- t information, implying evolving estimates of shock characteristics;
3. **Correlation surprises:** $\text{Cov}_0(D_t, S_t) \neq \text{Cov}_t(D_t, S_t)$, meaning initial estimates of shock co-movement prove incorrect.

Proof Sketch: Each condition creates wedges between ex-ante optimal policy (optimal given information at $t = 0$) and ex-post optimality (optimal given information at $t > 0$). For condition (1), the non-linear loss function $\omega(S_t)$

implies that the central bank's optimal policy at $t = 0$ depends on the *distribution* of S_t . If the perceived variance (or other moments of the distribution) of S_t changes as new information arrives, the ex-post optimal policy will differ from the ex-ante optimal plan, even if the underlying shock process remains the same. This is a mathematical consequence of Jensen's inequality if the loss function is convex in S_t . For condition (2), if the central bank's real-time estimate of the unobserved demand shock D_t evolves, the optimal policy response derived at $t = 0$ (based on initial estimates) will be suboptimal at $t > 0$ if the refined estimates differ significantly. This is a standard result in dynamic games with asymmetric information or learning. For condition (3), the initial policy setting would account for a specific expected correlation between D_t and S_t . If the realized correlation or the central bank's updated estimate of this correlation deviates from the initial expectation, the policy becomes suboptimal ex-post, leading to time inconsistency in the aggregate response. A formal demonstration, through deviations in first-order conditions or through a re-evaluation of the welfare function with updated information sets, is left for future work. $\square$

4.2 Welfare Analysis

Define the welfare loss from discretion as $\Delta W = \mathbb{E}[L^D] - \mathbb{E}[L^C]$ where L^D and L^C represent losses under discretion and commitment respectively.

Proposition 4 (Dual-Shock Welfare Loss): The welfare loss from discretion takes the schematic form:

$$\Delta W = \underbrace{\frac{\alpha \lambda_D^2 \phi_D^2 \sigma_D^2}{(1 - \beta) \mathcal{D}^2}}_{\text{Demand shock component}} + \underbrace{\frac{\alpha \lambda_S^2 \mathbb{E} \left[\left(\frac{\partial \omega(S_t)}{\partial S_t} \right)^2 \right] \sigma_S^2}{(1 - \beta) \mathcal{D}^2}}_{\text{Supply shock component}} + \text{cross terms} \quad (9)$$

where $\mathcal{D}$ stands for the denominator of the policy functions in equations (7) and (8), and the cross terms are not written out.

The key insight is that, in this expression, the supply shock component rises more than proportionally with supply shock volatility due to the non-

linear loss function $\omega(S_t)$, whereas the demand shock component is linear in σ_D^2 . This explains why traditional central banking frameworks, designed for stable economic environments, perform poorly during periods of high shock volatility.

4.3 Communication and Credibility

Central bank communication becomes crucial in dual-shock environments as private agents must form expectations over both inflation outcomes and the bank's ability to distinguish shock types.

Let $\theta_t \in [0, 1]$ represent private sector confidence in the central bank's shock identification ability. The modified Phillips curve becomes:

$$\pi_t = \alpha[\theta_t \pi_t^{e, informed} + (1 - \theta_t) \pi_t^{e, confused}] + \beta_D \max(D_t, 0) + \beta_S S_t + \varepsilon_t \quad (10)$$

where $\pi_t^{e, informed}$ would be expectations under perfect central bank credibility and identification, and $\pi_t^{e, confused}$ would be expectations formed with a less precise or biased understanding of shock compositions.

When θ_t is low (poor central bank credibility), inflation expectations become less anchored, amplifying the dynamic inconsistency problem and creating additional welfare costs from suboptimal expectation formation.

5 Policy Applications and Analysis: Fed Policy 2020-2024

5.1 Empirical Motivation and Stylized Facts

The Federal Reserve's experience during 2020-2024 offers a stylized illustration of our theoretical framework: the model's **qualitative predictions align with observed stylized facts** from this period. The period featured unprecedented overlap of massive supply and demand shocks, creating exactly the conditions our model predicts will generate enhanced dynamic inconsis-

tency problems.

Stylized Fact 1: Co-occurrence of Supply and Demand Shocks. Major supply disruptions included COVID-19 supply chain disruptions, the Ukraine war impact on energy and food prices, China’s zero-COVID policy effects, and shipping bottlenecks. Simultaneously, demand pressures arose from significant fiscal stimulus (e.g., American Rescue Plan), earlier pandemic relief packages, and massive sectoral reallocation from services to goods. The New York Fed’s Global Supply Chain Pressure Index peaked significantly while durable goods consumption surged, providing clear evidence of simultaneous and substantial supply and demand pressures that our model captures.

Stylized Fact 2: Initial Misidentification of Shocks. The Fed initially characterized inflation as “transitory,” attributing it primarily to supply-side bottlenecks. This aligns with a policy stance that, under our model, corresponds to an overweighting of supply shock implications and an underweighting or misidentification of concurrent demand pressures.

5.2 Policy Mistakes and Dynamic Inconsistency

The Fed’s “transitory” inflation narrative exemplifies dynamic inconsistency under dual shocks. Chair Powell’s August 2021 Jackson Hole speech described the spike in inflation as “so far largely the product of a relatively narrow group of goods and services that have been directly affected by the pandemic and the reopening of the economy” (Powell, 2021), essentially treating the situation as a pure supply shock requiring accommodation.

This assessment proved time-inconsistent as demand pressures intensified. Beginning in March 2022, the Fed implemented its fastest rate hiking cycle since the 1980s (525 basis points between March 2022 and July 2023), effectively treating the situation as a demand shock requiring aggressive tightening. Our theoretical framework, particularly Theorem 1, interprets this policy reversal as a consequence of evolving information and misidentification in a dual-shock environment.

Taylor Rule Analysis: Our framework offers one explanation for why

traditional Taylor rules provided poor guidance. Standard rules in 2021 suggested rates should have risen to 2-3% while the Fed maintained near-zero rates. Modified rules incorporating our dual-shock framework could, in principle, have prescribed earlier but more measured tightening, avoiding both the initial policy error and subsequent overcorrection; we do not compute such a rule path here.

5.3 Communication Challenges

The Fed’s communication difficulties illustrate our model’s predictions about credibility and expectation formation. The “transitory” messaging created forward guidance constraints that limited policy flexibility when new information emerged. Market-based inflation expectations rose substantially as private agents lost confidence in the Fed’s shock identification ability (our parameter θ_t declined).

Communication challenges were not confined to the United States. ECB analysis of its own monetary policy statements (Gardt et al., 2021) reports readability scores implying roughly 15-16 years of education to understand them before July 2021 and roughly 13-14 years afterwards. Even after that simplification, the statements remained demanding for the general public, which may limit the effectiveness of expectation management.

5.4 Welfare Costs

Our framework helps organize the discussion of the costs of policy mistakes. Core PCE inflation reached 5.6% in February 2022 (and again in September), 3.6 percentage points above the Fed’s 2% objective, which is defined for headline PCE, and stayed between 4.9% and 5.6% throughout 2022. Real wages declined substantially, particularly affecting lower-income households. The subsequent policy correction created financial stress, including banking sector strains in March 2023.

These outcomes are consistent with our theoretical prediction that welfare losses from discretionary policy increase substantially under dual shocks,

particularly when shock identification fails.

6 Conclusion and Policy Implications

This paper argues that dual-shock environments fundamentally alter the dynamic consistency properties of monetary policy. When central banks simultaneously face supply disruptions and demand pressures requiring conflicting optimal responses, traditional frameworks break down and welfare costs of discretionary policy increase substantially.

Our theoretical analysis yields several key policy implications. First, central banks should develop separate policy response coefficients for supply and demand shocks, in the spirit of the supply shock term in our modified Taylor rule. This requires improved real-time shock identification capabilities, potentially including sectoral inflation decompositions and supply chain monitoring systems.

Second, central bank communication becomes more complex but critical in dual-shock environments. Our framework suggests conditional forward guidance linking policy commitments to observable shock indicators rather than calendar dates could improve outcomes. Additionally, explicit acknowledgment of uncertainty and shock identification challenges may paradoxically improve credibility by setting appropriate expectations about policy flexibility.

Third, the enhanced welfare costs of discretionary policy under dual shocks strengthen the case for rule-based frameworks, but these rules must be modified to account for shock heterogeneity. Simple Taylor rules are likely to perform poorly when supply shocks are large or when shocks are misidentified.

The post-COVID period provides a suggestive illustration of these theoretical predictions. The Federal Reserve’s initial characterization of inflation as “transitory” followed by aggressive policy reversal illustrates the dynamic inconsistency problems our framework identifies. Similar patterns appeared in other advanced economies, suggesting these challenges reflect fundamental changes in economic structure rather than country-specific factors.

Future research should extend this framework to incorporate fiscal-monetary interactions, international spillovers, and financial stability considerations. The increasing frequency of global supply chain disruptions and geopolitical tensions suggests dual-shock environments may become the norm rather than the exception, making these theoretical insights increasingly relevant for practical monetary policy design.

Our framework provides a foundation for understanding and improving monetary policy in an era of increased economic fragmentation, supply chain vulnerability, and shock complexity. As traditional economic relationships continue evolving, theoretical models must adapt to address new sources of policy challenges while maintaining the fundamental insights of established frameworks like Kydland-Prescott dynamic inconsistency analysis.

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All remaining errors are my own.

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